

CO4 – AT1 DATA INTERPRETATION

Answered Questions

The answers below follow the wording, data, and evaluation requirements given in the uploaded question paper.

1. Real-Time Detection System

Question: A surveillance system uses the following techniques: Viola-Jones – 78% accuracy and Fast speed; YOLO – 92% accuracy and Moderate speed; Deep Learning – 95% accuracy and Slow speed. Compare the three techniques and evaluate which method should be selected for a surveillance system operating under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Answer: Viola-Jones is the fastest method, but its 78% accuracy is considerably lower. Deep Learning provides the highest accuracy at 95%, but its slow speed makes it unsuitable when latency is a strict requirement. YOLO provides 92% accuracy with moderate speed, giving a strong compromise between accuracy and response time.

Selection: YOLO. For a surveillance system with strict latency constraints, YOLO is the most appropriate choice because it maintains high accuracy while avoiding the slow response associated with Deep Learning. Viola-Jones may be selected only when extremely fast response is more important than detection accuracy, such as a resource-constrained deployment. If the deployment hardware can support higher computational cost and latency is relaxed, Deep Learning would be preferable for its 95% accuracy.

2. Robust Object Detection

Question: Compare the robustness of the three methods under normal, occluded, and low-light conditions. Evaluate which method provides the best overall performance and discuss the trade-offs involved.

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Answer: Deep Learning performs best in all three conditions: 95% under normal conditions, 85% under occlusion, and 80% under low-light conditions. YOLO ranks second with 90%, 75%, and 70%, while Viola-Jones has the lowest results at 80%, 60%, and 55%.

Overall selection: Deep Learning. It provides the strongest robustness and the smallest performance loss across difficult conditions. The trade-off is computational cost and slower processing, as indicated by the first question. YOLO is a practical compromise when real-time response is more important. Viola-Jones has the advantage of speed but sacrifices robustness, especially under occlusion and low light.

3. Optical Flow Analysis

Question: Two optical flow methods are available: Method A produces stable motion vectors but misses small motions. Method B detects fine motion but produces noisy vectors. Compare the two methods and evaluate which method is more suitable for motion tracking in dynamic scenes. Justify your answer based on accuracy, stability, noise sensitivity, and application requirements.

Answer: Method A has better stability because its motion vectors are stable, but it loses small or subtle motions. Method B has better sensitivity to fine motion, which can provide more detailed motion information, but its vectors are noisy and therefore less stable.

Selection: Method B for dynamic scenes where fine-motion accuracy is important, provided that noise can be controlled. In applications such as detailed motion analysis, tracking small movements is valuable, so Method B is more suitable if filtering or smoothing can reduce its noise. Method A is preferable when stable tracking is more important than detecting small movements—for example, applications where noisy vectors would cause unstable tracking. Therefore, the final choice depends on application requirements: fine-motion accuracy favors B, while stability and low noise favor A.

4. Model Generalization and Overfitting – 10 Marks

Question: Model A has 98% training accuracy and 70% testing accuracy. Model B has 90% training accuracy and 88% testing accuracy. Compare the two models and evaluate which model is more reliable for real-world deployment. Explain the generalization ability and overfitting risk of each model and justify your selection.

Model	Training Accuracy	Testing Accuracy	Train–Test Gap
A	98%	70%	28 percentage points
B	90%	88%	2 percentage points

Answer: Model A has very high training accuracy (98%) but much lower testing accuracy (70%), giving a 28-percentage-point gap. This indicates strong overfitting: the model performs very well on training data but does not generalize well to unseen data.

Model B has 90% training accuracy and 88% testing accuracy, with only a 2-percentage-point gap. Its training and testing performance are much closer, indicating better generalization and substantially lower overfitting risk.

Selection: Model B. It is more reliable for real-world deployment because real-world data is generally unseen during training. Although Model A achieves higher training accuracy, Model B provides much stronger evidence of generalization based on the given testing accuracy.

5. System Performance under Constraints

Question: Compare the three models and evaluate which model provides the best balance between accuracy, latency, and cost for a real-time application. Justify your choice considering practical deployment constraints.

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Answer: Model A has the lowest latency (50 ms) and lowest cost, but also the lowest accuracy (88%). Model B improves accuracy to 92% while keeping latency at 80 ms and cost at a moderate level. Model C has the highest accuracy (95%) but also the highest latency (120 ms) and high cost.

Selection: Model B. For a real-time application, Model B provides the best overall balance. It gains 4 percentage points of accuracy over Model A while adding only 30 ms of latency and moving from low to moderate cost. Model C gives only 3 additional percentage points of accuracy over Model B but adds 40 ms of latency and high cost. Therefore, unless the deployment specifically requires the highest possible accuracy and can tolerate greater latency and cost, Model B is the most practical choice.

Conclusion

Based strictly on the supplied data: YOLO is preferred for strict-latency surveillance; Deep Learning is strongest for robustness; Method B is preferred for dynamic scenes when fine motion is important and noise can be controlled; Model B has the best generalization; and Model B provides the best overall balance for the real-time system constraints.